

Calculus II Lesson 24

Taylor Series

Taylor Series Centered at c :

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n$$

$$f(x) = \frac{f(c)}{0!} (x-c)^0 + \frac{f'(c)}{1!} (x-c)^1 + \frac{f''(c)}{2!} (x-c)^2 + \frac{f'''(c)}{3!} (x-c)^3 + \frac{f^{(4)}(c)}{4!} (x-c)^4 + \dots + \frac{f^{(n)}(c)}{n!} (x-c)^n + \dots$$

1) Find Taylor series for the function $f(x) = e^{2x}$ centered at $c = 0$?

$$f(x) = e^{2x}$$

a) Find $f(0) =$ _____

b) Find $f'(0) =$ _____

c) Find $f''(0) =$ _____

d) Find $f'''(0) =$ _____

e) Find $f^{(4)}(0) =$ _____

f) Find $f^{(5)}(0) =$ _____

g) Find Taylor series for the function $f(x) = e^{2x}$ centered at $c = 0$? (multiple choice)

$$\text{Taylor Series Formula: } f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n$$

A) $f(x) = \sum_{n=0}^{\infty} \frac{4^n}{n!} (x)^n$

B) $f(x) = \sum_{n=0}^{\infty} \frac{3^n}{n!} (x)^n$

C) $f(x) = \sum_{n=0}^{\infty} \frac{2^n}{n!} (x)^n$

D) $f(x) = \sum_{n=0}^{\infty} \frac{1^n}{n!} (x)^n$

2) Find a Taylor series for the function $f(x) = \cos x$ centered at $c = \frac{\pi}{4}$?

a) Find $f(\pi/4) =$ _____?

b) Find $f'(\pi/4) =$ _____?

c) Find $f''(\pi/4) =$ _____?

d) Find $f'''(\pi/4) =$ _____?

e) Find $f^{(4)}(\pi/4) =$ _____?

f) Find $f^{(5)}(\pi/4) =$ _____?

g) Taylor series for the function $f(x) = \cos x$ centered at $c = \frac{\pi}{4}$? (multiple choice)

Taylor Series Formula: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n$

A) $f(x) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right) + \frac{\frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right)^2}{2!} - \frac{\frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right)^3}{3!} + \frac{\frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right)^4}{4!} - \frac{\frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right)^5}{5!} + \dots$

B) $f(x) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right) - \frac{\frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right)^2}{2!} + \frac{\frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right)^3}{3!} + \frac{\frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right)^4}{4!} - \frac{\frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right)^5}{5!} - \dots$

C) $f(x) = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right) + \frac{\frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right)^2}{2!} + \frac{\frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right)^3}{3!} - \frac{\frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right)^4}{4!} - \frac{\frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right)^5}{5!} + \dots$

3) Find Taylor series for the function $f(x) = \frac{1}{x}$ centered at $c = 1$?

a) Find $f(1) =$? _____

b) Find $f'(1) =$? _____

c) Find $f''(1) =$? _____

d) Find $f'''(1) =$? _____

e) Find $f^{(4)}(1) =$? _____

f) Find $f^{(5)}(1) =$? _____

g) Find Taylor series for the function $f(x) = \frac{1}{x}$ centered at $c = 1$? (multiple choice)

Taylor Series Formula: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x - c)^n$

A) $f(x) = \sum_{n=0}^{\infty} (x - 1)^n$

B) $f(x) = \sum_{n=0}^{\infty} (-1)^n (x + 1)^n$

C) $f(x) = \sum_{n=0}^{\infty} (-1)^n (x - 1)^n$

4) Find Taylor series for the function $f(x) = \ln x$ centered at $c = 1$.

a) Find $f(1) = \underline{\hspace{2cm}} ?$

b) Find $f'(1) = \underline{\hspace{2cm}} ?$

c) Find $f''(1) = \underline{\hspace{2cm}} ?$

d) Find $f'''(1) = \underline{\hspace{2cm}} ?$

e) Find $f^{(4)}(1) = \underline{\hspace{2cm}} ?$

f) Find $f^{(5)}(1) = \underline{\hspace{2cm}} ?$

g) Find Taylor series for the function $f(x) = \ln x$ centered at $c = 1$.

$$\text{Taylor Series Formula: } f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n$$

$$\text{A) } f(x) = \sum_{n=1}^{\infty} \frac{(x-1)^n}{n} \quad \text{B) } f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (x-1)^n}{n} \quad \text{C) } f(x) = \sum_{n=1}^{\infty} (-1)^{n+1} (x-1)^n$$

5) Find Taylor series for the function $f(x) = \sin 3x$ centered at $c = 0$.

a) Find $f(0) = \underline{\hspace{2cm}}?$

b) Find $f'(0) = \underline{\hspace{2cm}}?$

c) Find $f''(0) = \underline{\hspace{2cm}}?$

d) Find $f'''(0) = \underline{\hspace{2cm}}?$

e) Find $f^{(4)}(0) = \underline{\hspace{2cm}}?$

f) Find $f^{(5)}(0) = \underline{\hspace{2cm}}?$

g) Find Taylor series for the function $f(x) = \sin 3x$ centered at $c = 0$?

A) $f(x) = \frac{3}{1!}(x)^1 - \frac{3}{3!}(x)^3 + \frac{3}{5!}(x)^5 - \frac{3}{7!}(x)^7 + \dots$

B) $f(x) = \frac{1}{1!}(x)^1 - \frac{1}{3!}(x)^3 + \frac{1}{5!}(x)^5 - \frac{1}{7!}(x)^7 + \dots$

C) $f(x) = \frac{3^1}{1!}(x)^1 - \frac{3^3}{3!}(x)^3 + \frac{3^5}{5!}(x)^5 - \frac{3^7}{7!}(x)^7 + \dots$

